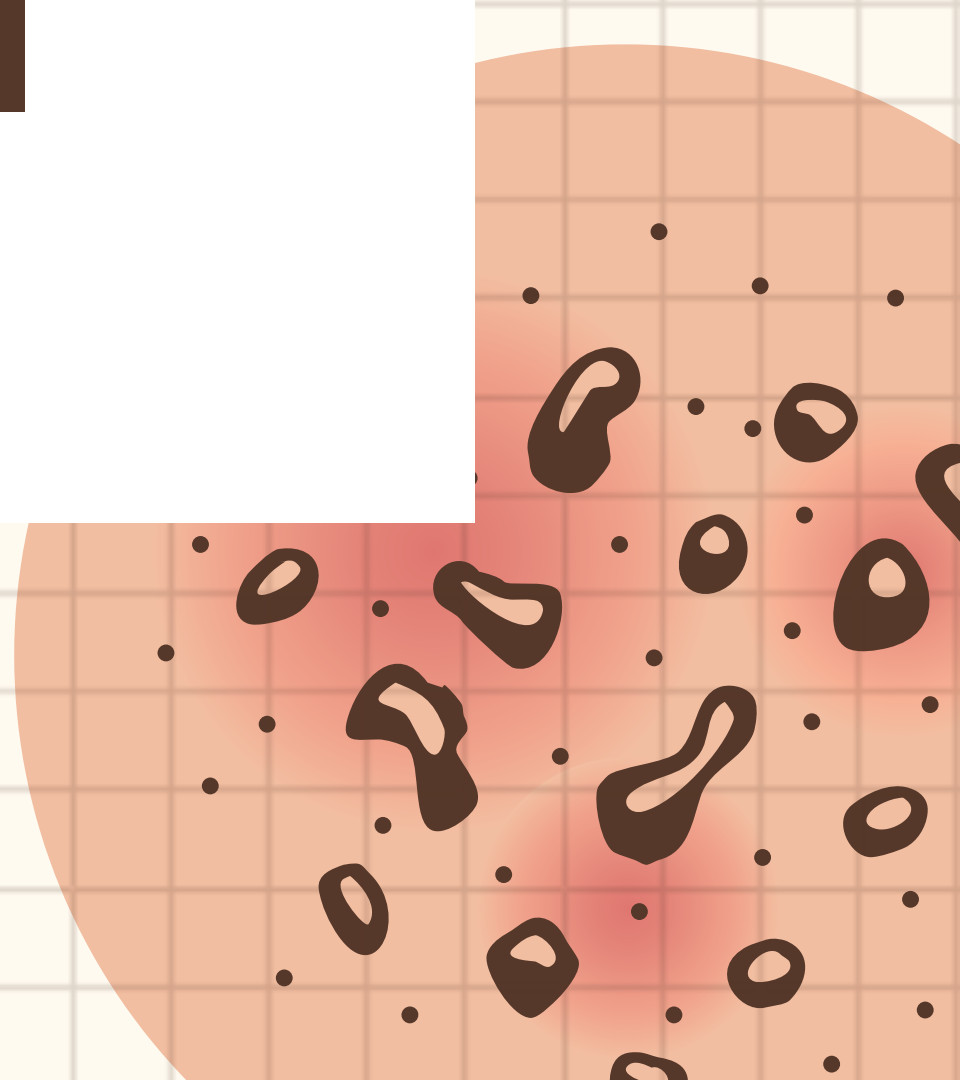




Skin lesion

Multimodal XAI

Team 15



Problem Statement

Challenges

- Melanoma is highly dangerous if detected late
- Dermoscopic lesions show subtle visual differences
- High intra-class similarity
- Irregular lesion borders and pigmentation
- Traditional systems ignore clinical metadata

Our Goal

Build explainable multimodal AI systems that:

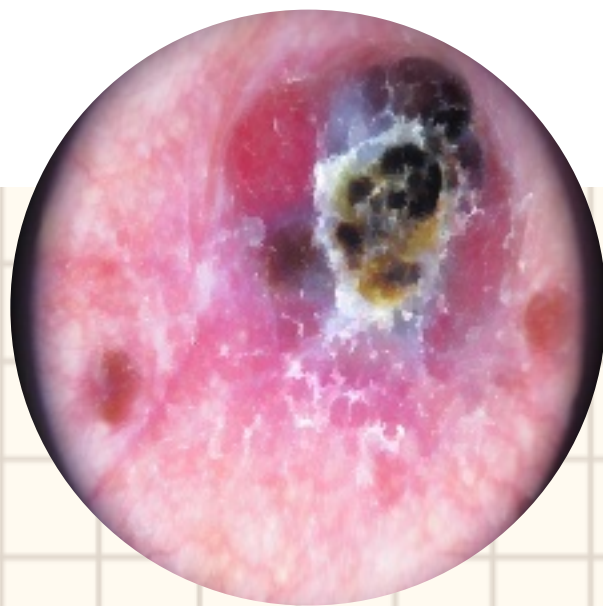
- Classify melanoma vs non-melanoma
- Combine image + metadata reasoning
- Provide interpretable explanations



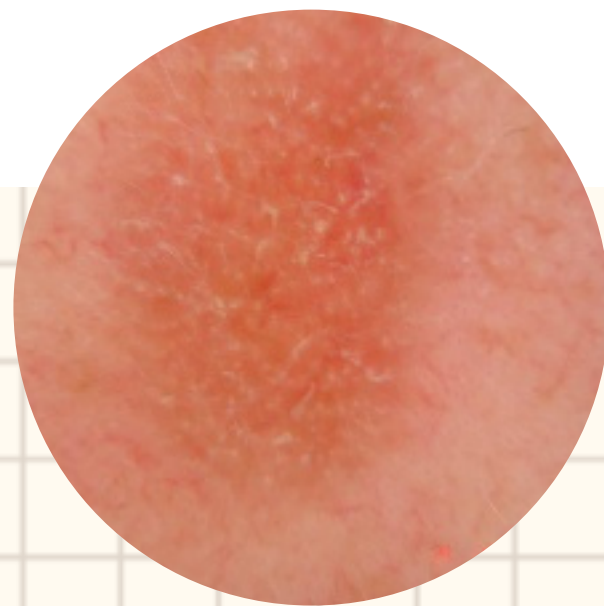
Dataset Overview

ISIC 2017 / ISIC-DICM-17K style

- Binary classification (Melanoma, Non-Melanoma)
- Modalities: Dermoscopic Images, Clinical Metadata



Melanoma

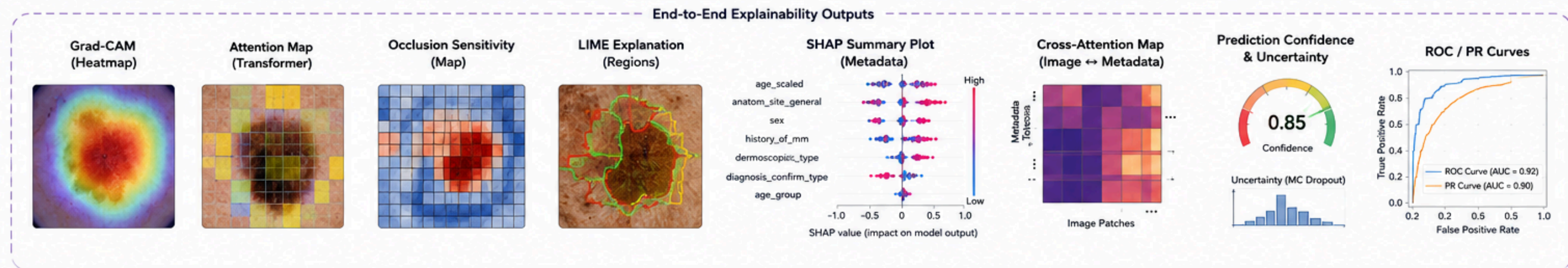
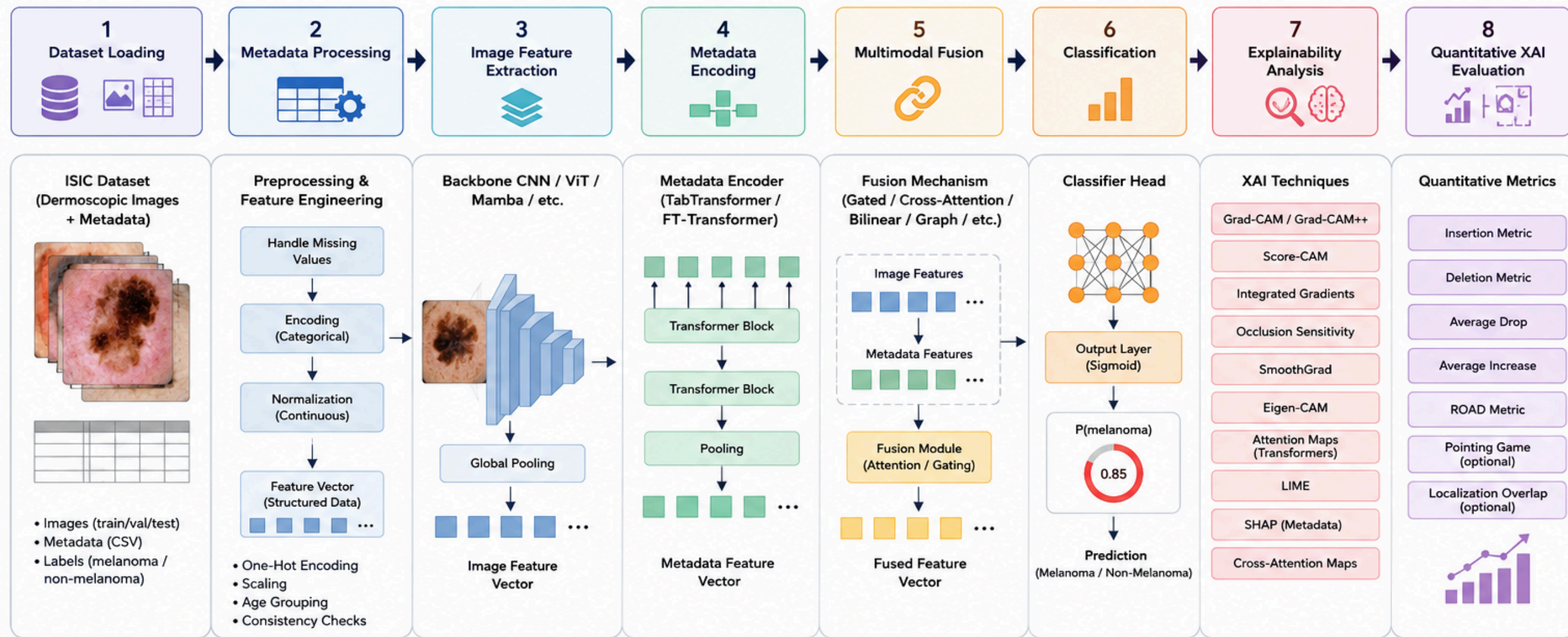


Non-Melanoma

Metadata Features

Age, Sex, Anatomical Site, Dermoscopic Type, History of Melanoma, Diagnosis Confirmation Type

Complete Multimodal Pipeline



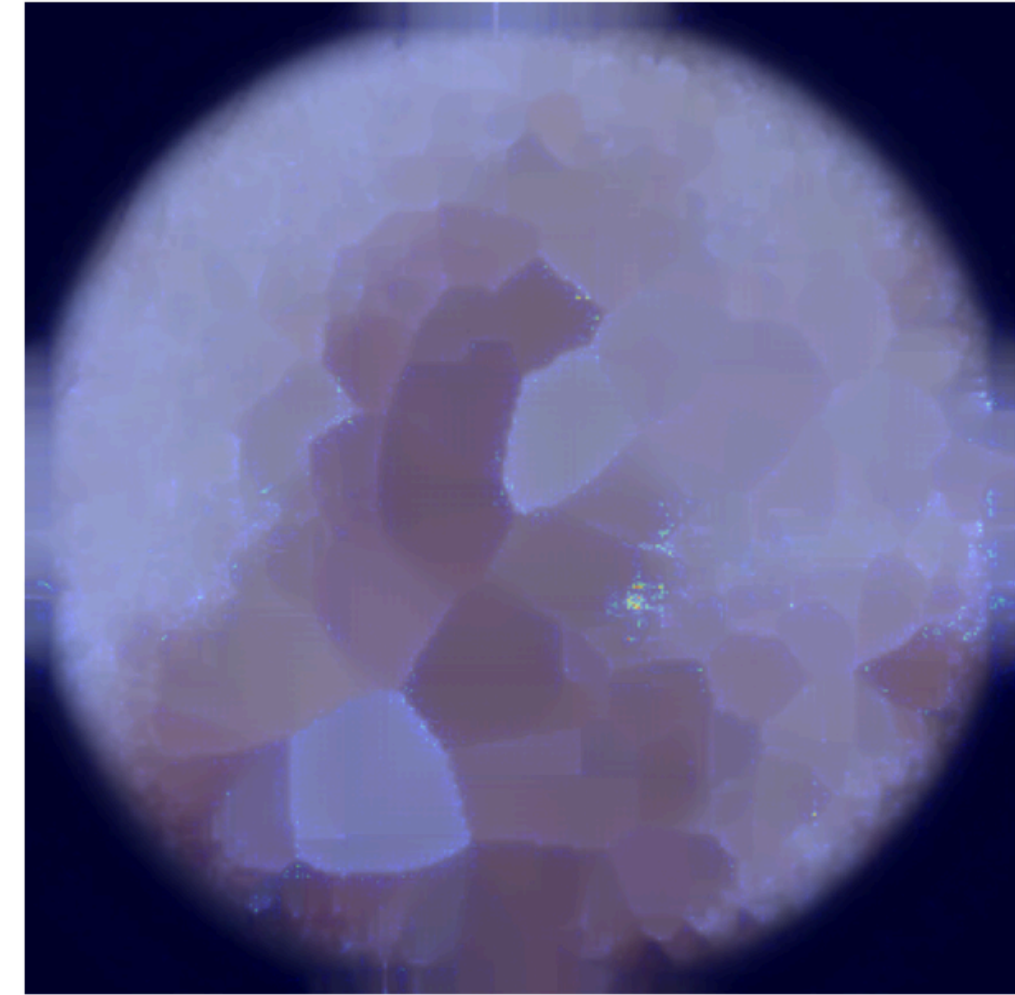
Implemented Deep Learning Architectures

Member	Models
Ahmed	1) DaViT-Tiny + Gated Fusion 2) Vision Mamba + Metadata Gated Fusion 3) BioViL-T + FT-Transformer + Lesion-Conditioned Sparse Attention 4) DINOv2-ViT-S/16 + TabTransformer + Cross-Attention
Abdelrahman	1) UNet-Encoder Multimodal Classifier + Metadata Cross-Attention 2) Swin Transformer V2 + Graph Attention Fusion 3) ConvNeXt + FT-Transformer + Bilinear Fusion 4) CLIP-style Contrastive Multimodal Model
Jana	1) MaxViT-T + Gated Fusion 2) EfficientNetV2-S + TabTransformer + Cross-Attention 3) Swin Transformer V2 + TabTransformer + Cross-Attention 4) SegFormer-B5 + TabTransformer + Lesion-Aware Cross-Attention
Malak	1) Graph Neural Network Fusion Model 2) ConvNeXt-Tiny + Bidirectional Cross-Attention 3) Hybrid ViT-MLP Late Fusion 4) ViT Early Fusion

XAI Techniques

- Grad-CAM
- Grad-CAM++
- Score-CAM
- Eigen-CAM
- LIME
- SmoothGrad

ConvNeXt + FT-Transformer + Bilinear Fusion | true=1 | melanoma prob=0.987 | Final model



Metadata & Multimodal XAI

SHAP : Global + local explanations

DeepSHAP : Deep-learning optimized SHAP for multimodel explanations

PDP Plots : Visualizes feature influence trends

- Cross-attention visualization
- Modality importance analysis

is considered as Quantitative XAI

Multimodal Fusion Techniques

Gated Fusion :

Learns modality importance dynamically

Cross-Attention Fusion :

Deep-learning optimized SHAP for multimodel explanations

Bidirectional Cross-Attention :

Two-way modality interaction

Bilinear Fusion :

- Captures nonlinear interactions

Training & Evaluation

- Optimization
 - · Early Stopping
 - AdamW Optimizer
- Evaluation Metrics
 - Recall
 - F1-score
 - · ROC-AUC
- Regularization
 - · Stochastic Depth
 - · Label Smoothing
 - Dropout

Results Overview

Best Performing Models:

- Swin Transformer V2 + TabTransformer + Cross-Attention
- Vision Mamba + Metadata Gated Fusion
- SegFormer-B5 + Lesion-Aware Cross-Attention
- ConvNeXt + FT-Transformer + Bilinear Fusion
- Multimodal fusion improved performance significantly
- Metadata enhanced classification quality
- Transformer models captured global lesion structure
- CNN-based models captured fine lesion textures effectively

Error Analysis

- Failure Cases & Limitations :
 - Small low-contrast lesions
 - Highly irregular pigmentation
 - Artifact-heavy images
 - Border ambiguity
- Some models focused on surrounding artifacts
- Difficult lesions produced uncertain predictions
- Metadata improved difficult-case classification

Conclusion

- Implemented multiple multimodal architectures with advanced XAI techniques
- Compared modern fusion strategies
- Multimodal learning improved melanoma detection
- Cross-attention fusion produced richer interactions
- Transformer-based models captured global lesion structure



Future Work

- Larger datasets
- Segmentation-guided learning
- Clinical deployment
- Real-time explainability systems
- Higher computational power





Thank you

for listening!

Questions?